

Project Initialization and Planning Phase

Date	16 June 2025
Team ID	SWTXT_20251747173488
Project Name	Productivity Prediction Web App
Maximum Marks	3 Marks

Customer Problem Statement

Saurabh Yadav
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Introduction The Productivity Prediction Web App tackles key challenges faced by garment industry managers and supervisors in forecasting worker productivity. Using machine learning models (Linear Regression, Random Forest, and XGBoost) trained on the "Productivity Prediction of Garment Employees" dataset, the app provides predictions with a Random Forest R^2 score of 0.547. The following problem statements outline the customers' challenges, obstacles, and emotions, guiding the development of a data-driven solution.

Who: Garment industry managers and production supervisors responsible for optimizing workforce performance.

What: These managers struggle to accurately predict worker productivity due to complex factors such as varying team sizes, overtime hours, and production targets, relying on manual estimates or outdated methods.

Why: Inaccurate productivity predictions lead to inefficient resource allocation, missed production targets, and reduced profitability, causing frustration and operational bottlenecks.

Desired Outcome: Managers need a reliable, data-driven tool that leverages machine learning to forecast worker productivity based on historical data, enabling informed decisions to enhance efficiency and profitability.

Solution Context: The Productivity Prediction Web App addresses this challenge by using machine learning models (Linear Regression, Random Forest, and XGBoost) trained on the "Productivity Prediction of Garment Employees" dataset. With a Random Forest model achieving an R^2 score of 0.547, the app processes inputs like standard minute value (SMV), work in progress (WIP), and team size to deliver accurate productivity predictions, empowering managers to optimize operations.

Example:

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Garment factory production managers	Achieve accurate production targets daily	Manual productivity predictions are unreliable	Complex variables like overtime and team dynamics are hard to analyze manually	Stressed about meeting deadlines and goals
PS-2	Garment industry supervisors	Plan workforce	Lack a tool to predict	Historical data isn't leveraged to	Overwhelmed by inefficient scheduling decisions

		schedules efficiently	team productivity accurately	account for factors like SMV or incentives	
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